Source

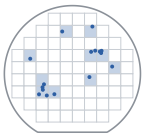
## Semiconductor yield engineering

Target

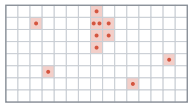
## Supply-chain early warning

**Inline inspection**

defects per die on wafer maps



**Detection**



**Event screening**

shock events per item-period

**Negative-binomial yield**

$Y = (1 + AD_0/\alpha)^{-\alpha}$

**Model**

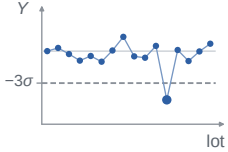
| | | | |
|---|---|---|---|
| $D_0$ | ↔ | $R$ | intensity |
| $A$ | ↔ | $C$ | exposure |
| $\alpha$ | ↔ | $\alpha'$ | dispersion |
| $Y$ | ↔ | SRI | reliability |

**Mixed-Poisson reliability**

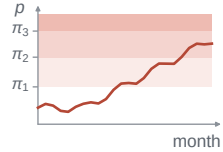
$\mathrm{SRI} = (1 + RC\Delta/\alpha')^{-\alpha'}$

**Disposition limits**

$\sigma$-limits on yield and scrap




**Alarm**




**Cost-based alarm**

four levels at thresholds $\pi_k$